McGill University — Desautels Faculty of Management

MGSC 670 - Revenue Management

Group Assignment 2:

Case Study - Pricing Analytics for Movie Theatres

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1 Introduction

When was the last time you saw a movie in theatres? And the time before that? Globally, movie theatres or box office sales revenue grew slowly and steadily from $25.4 billion in 2007 to $42.5 billion in 2019. [Statista, 2023] The covid pandemic disrupted revenue growth when box office revenue dropped from $42.5 billion in 2019 to $12.4 billion in 2020, about a 70% decrease globally and 80% in the United States and Canada. [Gitnux, 2023] The industry still needs to recover from the impact of the pandemic, as shown by Cineplex 2022 Year Financial Results (cf. Figure 1, 2 & 3). Yet, movie production remained significant, with about 40,613 movie screens in the U.S. in 2020.

Curiously movie theatre ticket sales, or the number of units sold in North America, have declined from 2000 to 2019. The number of tickets sold went from 1.5 billion in 2000 to 1.25 billion in 2019, indicating that fewer people attended the movies throughout this period. See Figure 4. [Nash Information Services, 2023] Ticket revenue increased because of ticket prices. Prices increased in part from features such as 3D, VIP or other premium offerings - Cineplex discloses a growing number of VIP, D-BOX and Recliner auditoriums as well as premium screens over the last two years. The majority of these are 3D or offer premium features demonstrating their commitment to more features. (cf. Figure 5: Cineplex Theatres, screens and premium offerings in the last eight quarters). Based on the slow revenue and decline of movie theatre attendance, the pandemic may have exacerbated the issues instead of causing them.

Movie theatres are slowly recovering; however, even with meme stock cash inflows, the largest grossing movie of all time and other factors, theatres have yet to establish market innovations to define their new baseline and incentive people to the theatres. To complicate the situation, streaming services erode movie theatres' demand. With restrictions lifted and more movies being released, theatres have an opportunity to address the slowing of attendance. The following study investigates pricing innovations that could stimulate theatre ticket sales further.

2 Background

The US movie industry globally produces over $100 billion USD, with approximately $42 billion coming from box office sales before the COVID pandemic. [Escandon, 2020] This represents a massive industry with millions of views for each movie. Movies generate revenue that is split between the producers and the theatre companies. A traditional split is 40% for the theatre company, although some arrangements allow these companies to recover 60% weeks after the initial showing of the movie. Theatre companies rely on sales of concessions, advertising, and ticket sales which generally make up 33%, 1% and 62% of revenue, respectively. [Seiler, 2023]

Between 1995 and 2019, revenue increased despite an estimated 16% decrease in attendance in North America. [Nash Information Services, 2023] The pandemic caused an estimated 53% decrease in sales. [Nash Information Services, 2023] During the pre-pandemic (1995-2019) era, ticket prices drove industry growth. Movie ticket prices grew at 4% annually, beating out the 2% target inflation. [Nash Information Services, 2023] (c.f. Figure 6). Part of this is directly attributable to higher-end services, like VIP, 3D and others. For example, "The Lion King" was a big hit in 2019 and became North America's most profitable 3D movie, with a box office revenue of $543.6 million. [Gitnux, 2023] China took advantage of the growing popularity of 3D movies and had 1,500 3D movie screens in 2021. Pathé developed a premium offer in France with sofas or comfortable long chairs with pillows, USB plugs and other gadgets. These premium features increase prices and, by extension, movie revenue.

Despite new features and higher ticket prices, the revenue growth has slowed down. Declining attendance is partially explained by substitute products such as streaming, and theatre companies' complacency with movie pricing. Other countries have strategic movie pricing like small group pricing, student pricing, or discounts after several weeks of the movie release. For instance, in Taiwan, several theatres known as 'second rounds' show movies for a discount after their initial release. Other theatres in Germany are implementing dynamic pricing.[Hazelton, 2022]Price differentiation models help decrease the deadweight loss produced from demand at lower prices and increase the profitability of movie tickets. (c.f. Figure 7 for a segmentation illustration). Adapting prices to different audiences would increase the volume of ticket sales. Moreover,

the theatre may further boost revenue through complementary effects of food and beverage purchases.

Theatres have tried to innovate with more luxurious offerings, subscription clubs, and room rentals for various events. However, they have yet to attempt to innovate pricing models to adapt to market demands.

3 Proposed Changes

Today, in Canada, prices for movies are primarily static. Some price differences do exist based on the features (3D, VIP) or vendor (Costco) but have been insufficient to capture additional ticket volume. We propose three different pricing options and their potential effects to address the dynamic pricing gaps within the cinema. The first option is small group pricing; the second is a markdown strategy like the fashion industry, and finally, market segmentation. These pricing options address market segments and strategies that are complementary to each other.

3.1 Group Pricing

A common strategy in North America is to offer special pricing for large groups. Large groups change based on the definition but are generally around 8-10 or more people. Group rates generally make up 20% of total demand. [Fabbro, 2020] Some studies point to the efficacy of increased revenue by adjusting the pricing based on intrinsic demand and supply. [Yunmei (Mabel) Bai, 2021] Marriot, for instance, optimized its group discount model and saw a profit increase of 1.1 percent. [Hormby et al., 2010] Group rates increase revenue and offer a risk reduction as the volume of sales increases. Furthermore, group pricing positively affects word of mouth and overall sales, demonstrating its value beyond revenue. [Bollinger et al., 2016] Almost all industries offer these rates as group rates because it increases profits. While movie theatres offer large group rates for events, they do not offer any small (4 people) or medium-sized (6 people) group discounts. Group discounts integrated within the pricing scheme and POS systems would encourage more groups and thus volume for movies. Customizing the size

of group discounts, specifically for smaller groups, is beneficial for increasing revenues. Small group discounts incentivize groups of friends to go together more often while limiting the consumer's psychological risk of paying for a large group. Creating a group pricing scheme for smaller groups would benefit the consumer and increase the profit of movie theatres through additional volume.

3.2 Markdown Pricing

Furthermore, the typical approach to market saturation for any movie is to decrease that movie's show times. Despite consolidating the demand from multiple screenings to one for a movie, few go to see the movie. Demand consolidation does not work in this industry. Therefore, with other operations remaining the same, we would like to introduce markdown pricing to expand the demand. Markdown pricing debuted in the theatre industry in the mid-20th century in a different fashion. Second-run theatres offer movies released in the first-run theatres yet at a lower price. Second-run theatres often show the movies weeks after their debut. However, this markdown strategy has risks. Numerous second-run theatres have faced closure or been forced out of operation due to the shortened delay between theatrical and home video releases. Inspired by this mechanism, we propose to include markdown pricing on long-running movies in the first-run theatres. According to the box office history dataset collected, in 2020 to 2022, customers tend to go to the movies they are interested in within the first two to three weeks after the movies are released. (c.f. Figure 8) In this case, adopting markdowns after that will allow to attract more customers. Even though the individual margin is lower with markdowns, this strategy could allow theatres to increase overall revenue by increased sales since the cost is almost fixed for one show time regardless of the number of customers. Additionally, revenues from beverage and snack bundles will increase with this strategy as the audience increases. Compared to the existing method of reducing show times when the demand is saturated, introducing markdown shifts the target to customers with a lower budget and therefore enlarges the market. At the same time, this approach can generate higher customer satisfaction as it improves their perceived value compared to paying the full price for the same movie.

3.3 Market Segmentation

The last pilar of our study is customer segmentation. Market segmentation is a crucial revenue management technique that cinemas can employ to enhance profitability. Student discounts are a common segmentation strategy across many industries – among which the cultural industry – that can reinforce customer retention and foster attractiveness and new business. One study found that 79% of students in the UK opted for a new brand because of discounts. [Cronin, 2021] Another study found that most students would feel loyal to brands based on discounts. [Review, 2021] This demonstrates students' sensitivity to pricing and willingness to purchase based on discounts. Theatres that do not currently offer student discounts in North American should cater to price-sensitive students as studies have demonstrated that movie goers are price elastic, with elasticities between 3 and 1.5. [De Roos and McKenzie, 2014] Segmentation based on student groups is expected to be particularly effective towards increased demand and loyalty to the brand. Surveys demonstrate that a 10% discount would convince 32% of students to purchase. [Hansen, 2012] Using this as a baseline, the analysis will explore how adjusting prices based on student elasticity can boost revenue. Cinemas will increase the volume of demand and specifically their students base, which will positively impact food services and ticket sales.

4 Analysis of Changes

To illustrate the impact of market segmentation through small group pricing, markdown strategies and student pricing, we will demonstrate how adjusting prices based on elasticity values can increase revenue. Elasticity refers to the responsiveness of demand to changes in price. In our case, a negative elasticity indicates that as prices increase, demand decreases, representing a negative relationship. An elasticity value greater than 1 indicates high price sensitivity, where a small price change can lead to a significant demand shift. Conversely, an elasticity value less than 1 suggests low price sensitivity, where substantial price changes have a more negligible effect on demand. For our analysis, we will consider both discounts and price increases for different market segments.

4.1 Simulation Setup

To conduct this analysis, we utilized a global box office performance dataset. [Mojo, 2023] The data set breaks down the daily ticket sales per movie. From this dataset, we can generate simulations of different strategies and the demand and price elasticity (c.f. Figure 9). Elasticity is important to understand assumptions and optimal pricing for consumers. As we move away from an elasticity value of 1, there is potential to adjust pricing. If a segment exhibits high elasticity, such as the student segment, it would be beneficial to offer them a discount to stimulate demand. Conversely, increasing prices could be advantageous for segments with low elasticity, such as regular customers wanting to purchase a ticket to the cinema. Based on our estimations of the elasticity of each segment, and by tailoring pricing strategies to specific market segments, cinemas can effectively capture additional revenue opportunities. The elasticity of customers against the revenue generation and optimal price point is plotted to allow informed estimates and assumptions on the impact of the chosen approaches. (c.f. Figure 9)

The dataset allowed several simulations, one for each proposed strategy. Using the dataset, we modelled the baseline, then a group pricing approach, next a markdown scheme, and then a student pricing model. The dataset gave us access to the box office sales. However, including a scenario which includes food purchases is necessary. Food, especially popcorn, is a lifeblood for many theatres that use it to be profit positive. [Crockett, 2019] Using a breakdown of revenue sources, we estimate that 1/3 of all revenue generated comes from food and concessions. Therefore, each strategy will include food and non-food considerations.

4.2 Small Group Pricing

First, the small group pricing. We assume that group sales must uphold the maximum sales volume possible. The maximum sales volume equals the highest observed sales within the dataset. We will also assume that a 10% discount in small group pricing leads to a flat 15% increase in demand with a 10% demand cannibalization effect. To normalize the data across all movies, we used a baseline of the first-day sales as a 100% sales indication. Most movies have a high sell-through rate within the first days, which gradually lowers. We found a cumulative

revenue of 1,309% against a baseline of 1,280%. Or 39% cumulative revenue increase over 50 days or a 3% increase in overall revenue. The increase in revenue is strongest after 17 days, or the typical high-demand period. This strategy is effective for additional demand and revenue. The model with food included shows similar trends with slightly higher revenue.

4.3 Markdown Pricing

Second is the markdown strategy. We will assume for this strategy that there are two markdown prices. The second-run pricing after 14 days is a 10% reduction in price, which we assume to have a 15% increase in demand. After 30 days, the price drops by 20%, with an assumed increase in demand of 20%. Similar assumptions for the sales indication of 100% of the first day apply. The markdown strategy for non-food consideration remains unchanged compared to the baseline. Even assuming a markdown of 10% would increase sales demand by 20% producing small variations in price. (c.f. Figure 10 and Figure 15) With food, similar to group pricing, we simulated an overall revue of 3% increase. This strategy could be more optimal compared to the small group pricing.

4.4 Student Pricing

Third, the student pricing model assumes a 25% decrease in price for a 25% demand increase with cannibalization effects of 15% on the original volume. The simulations demonstrate a cumulative revenue of 1,474% against the baseline of 1,280%. This represents an increase of 194 % in the cumulative amount or 15% overall higher revenues by day 50. The food meanwhile shows slightly higher numbers. The differentiation begins at day 10 and increases the revenue beyond this date. The student pricing option is an effective one for increasing the revenue of both ticket sales and concessions.

4.5 Combined Model

Finally, a combined model demonstrates the combined power of all models by accounting for the cannibalization and cross-pricing group effects. Based on the simulation of these prices, the cumulative revenue increase is 100% or a 7.8% overall revenue lift. The trend with food generates

higher total revenue and an overall increase in revenue of 11.7% by day 50. Therefore, the total revenue increase is smaller than the student discount but higher than the other strategies. It is well worth considering all strategies for movie theatres as their box office and food revenues will increase accordingly.

5 Conclusion & Extensions

Movie theatre's lack of innovation in pricing has led to a decline in viewership over the last 30 years. The pandemic presents a unique opportunity to revisit the industry assumptions on what consumers want. Filling a movie theatre is conceptually like filling an aircraft. There is limited space, some premium features and no salvage value on unused seats. Unlike the airline industry, movies use a 'one size fits most' approach to pricing for seats. This report's innovative approaches for pricing provide benefits and demonstrate the potential of thinking outside the box. The proposed incentives include group pricing, markdowns on less popular movies, and customer segregation for students, aiming a better utilization and revenue management.

The first proposed approach, group rates, provides customer incentives to go to movies in larger groups, thereby mitigating the risk of decreased sales with increased volume. A smaller-size group discounts can increase the audience and potentially, reinforce the herd mentality thereby increasing the popularity of movies; either case will result in better revenue. Groups are only sometimes available for movies so what if ticket prices dropped as movies become less popular? Considering the movie Barbie with Margot Robbie and Ryan Gosling coming out next July, it will likely be popular in the first few weeks and attract regular cinemagoers or students on summer break. After the initial wave of viewers, and into the start of school the popularity will decrease soon as the next big movie takes over the theatres. The decrease in attendance is exponential. However, we estimate that markdown pricing smooths this decrease, and maximizes revenue by attracting more demand at the end of life of movies in theatres. Much like Via rail offers discounts on Tuesdays or restaurants offer discounts at 'happy hour'. Last, we believe a more granular market segmentation of cinema's audience will help adequately address the demand with a broader range of choices. Museums fee schedule can be a good

example of customer segmentation in the cultural industry. With different offers depending on each segment, movie theaters can increase traffic and take advantage of demand elasticity by offering student pricing. We estimate that a segmented pricing scheme results in a win-win scenario: more profit for theatres with more satisfaction, positive word of mount and loyalty for the audience due to perceived benefits. For further details on implementation and business case please see the business case in the appendix.

In addition to the proposed approaches, we conclude with other levers that could be investigated for future extension. Seat-location-based pricing could be another path to exploit to tackle customers' frustration when they end up in the bottom right corner of the room. This policy could segregate audiences with different needs. For example, some customers might focus on movie content and prefer seats at lower prices even though the stereo quality might not be the best while some value the overall theatre experience more and will not compromise on seat locations. A second extension is a more intuitive booking interface that will improve the customer experience. Current portals do not offer customized experience for users. Therefore, a web portal for personal accounts, that suggests movies, remembers preferences and sends notifications to allow customers to book in advance would prove beneficial. Other features could include choosing a cancellation fare, pick seats, or preorder foodservices, while offering users a customized and smooth experience. In addition, such a platform would be a significant source of insights for sales forecasting, data-driven marketing – bundles, A/B testing, traffic and click-through rates – and revenue optimization of each pillar of the revenue mix – food services amounting for up to one third of the profits of Cineplex in 2022 (cf. Figure 3: Cineplex revenue mix comparison [Cineplex, 2022]).

This report proposes different strategies to address the stagnant price strategies in the cinema industry. Many of these approaches share similarities among different industries such as airlines and hotels. These similarities exemplify how movie theatres can adapt their business structures and adopt various revenue management techniques in their operations. These strategies are feasible, easily implementable and necessary for customer retention and to mitigate the impact of online streaming services on the box office.

6 Appendix

6.1 Figures

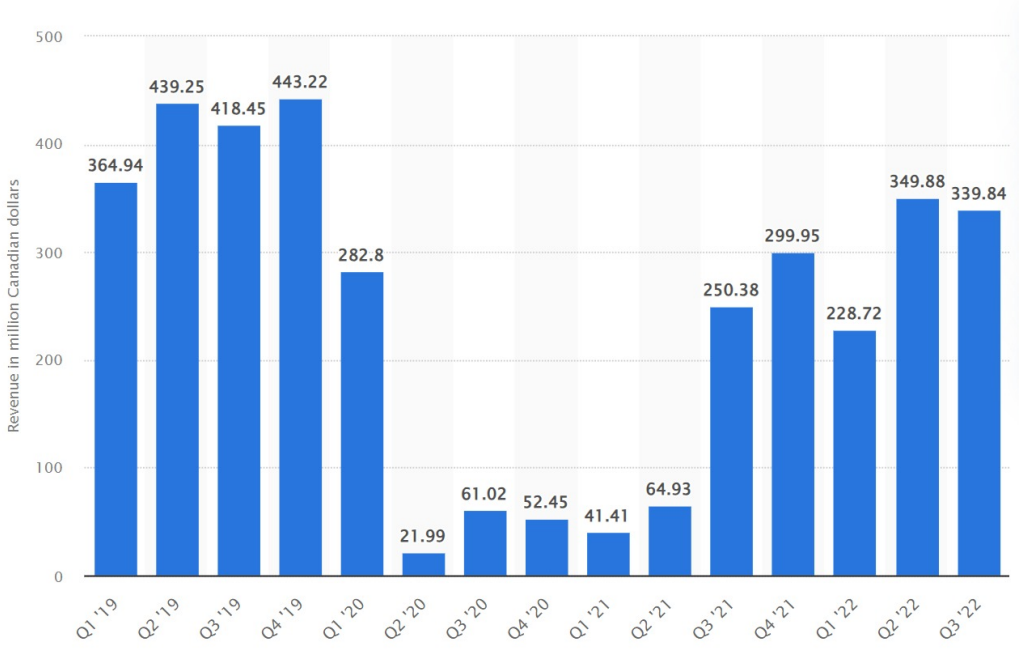


Figure 1: Quarterly revenue generated by Cineplex Inc. from 1st quarter 2019 to 3rd quarter 2022 (in million Canadian dollars) [Starista, 2022]

Month	2019 Box Office (i)	2022 / 2023 Box Office (i)	2022 / 2023 as a Percentage of 2019
October	$54,528	$33,907	62 %
November	$52,314	$37,981	73 %
December	$74,947	$48,359	65 %
Q4 Total	**$181,789**	**$120,248**	**66 %**
January	$52,034	$45,744	88 %
(i) Amounts are in thousands of dollars.			

Figure 2: Cineplex fourth quarter comparison 2019-2022 [Cineplex, 2023a]

Revenue mix % by period	2022	2021	2020	2019	2018
Box office	**36.4 %**	36.0 %	31.8 %	42.4 %	44.9 %
Food service	**30.1 %**	28.5 %	26.0 %	29.0 %	29.5 %
Media	**8.8 %**	9.9 %	15.6 %	11.8 %	10.1 %
Amusement	**19.4 %**	20.5 %	18.6 %	13.7 %	12.8 %
Other	**5.3 %**	5.1 %	8.0 %	3.1 %	2.7 %
Total	**100.0 %**	100.0 %	100.0 %	100.0 %	100.0 %

Figure 3: Cineplex revenue mix comparison over years – 2023 Cineplex Annual Information Form [Cineplex, 2022].

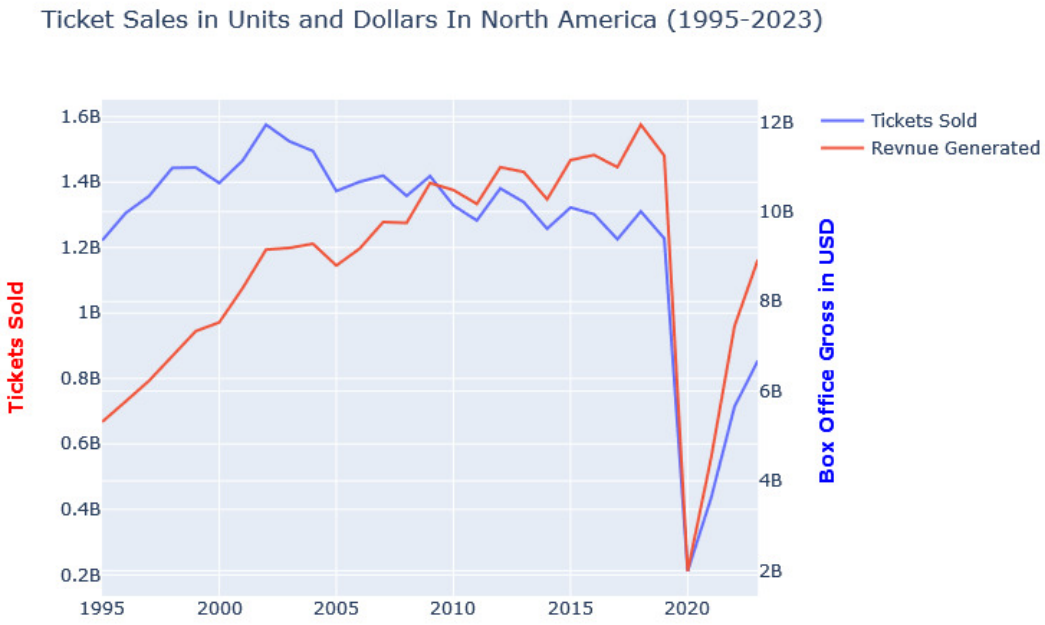


Figure 4: Tickets sold in units (blue) vs. Revenue generated (red) from 1995 -2023 for movie box office sales.

Cineplex - Theatres, screens and premium offerings in the last eight quarters								
	2022				2021			
	Q4	Q3	Q2	Q1	Q4	Q3	Q2	Q1
Theatres	158	158	159	159	160	161	160	161
Screens	1,637	1,637	1,640	1,640	1,652	1,656	1,651	1,657
3D Digital Screens	809	809	809	810	815	816	816	816
UltraAVX Screens	95	94	94	94	94	94	94	94
IMAX Screens	25	25	25	24	25	25	25	25
VIP Auditoriums	99	99	99	99	99	94	89	84
D-BOX Auditoriums	100	98	98	98	98	98	98	98
Recliner Auditoriums	273	267	267	267	267	262	258	253
Other Screens	27	23	22	22	22	19	19	19

Figure 5: Cineplex Theatres, screens and premium offerings in the last eight quarters – 2023 Cineplex Annual Information Form [Cineplex, 2022]

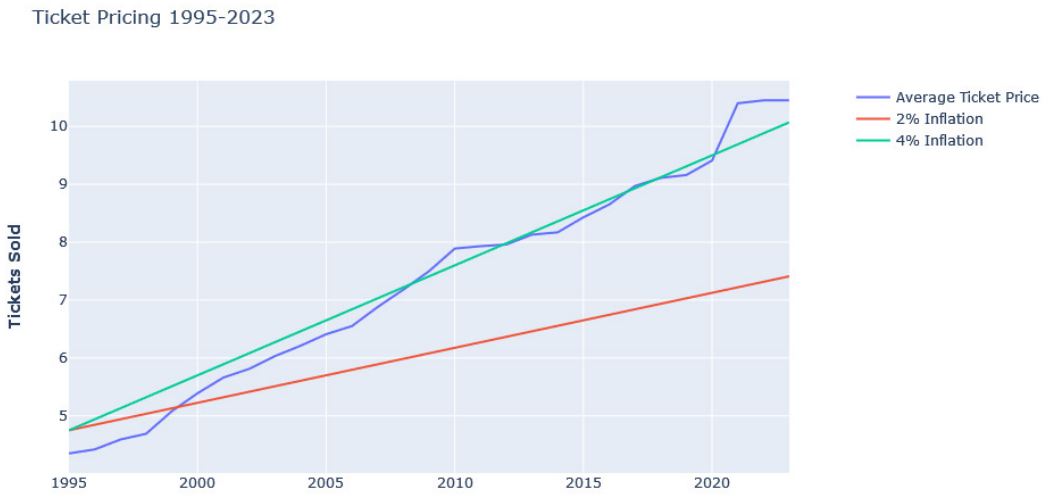


Figure 6: Ticket Price at box office 1995-2023. Increase of 3% compounded annually

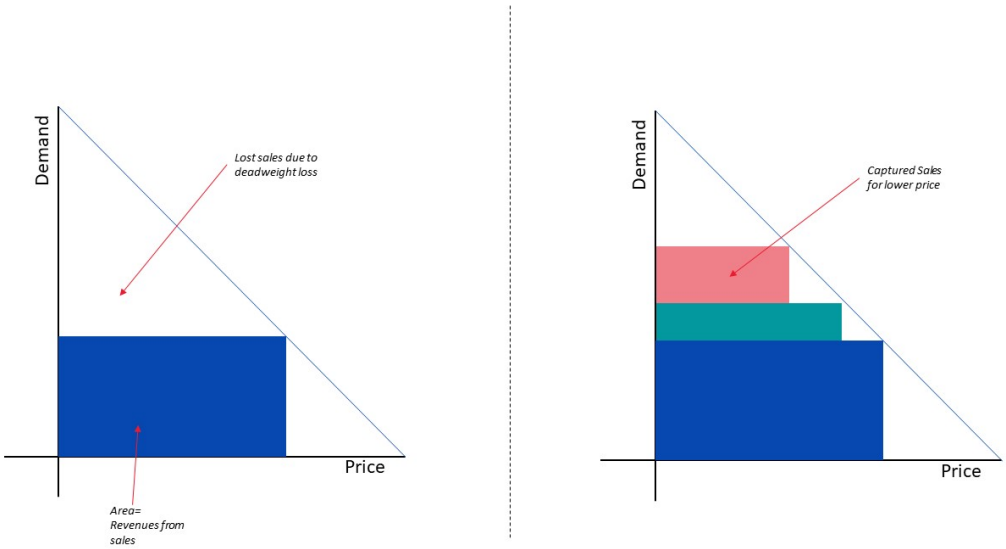


Figure 7: Price Segmentation Visual Aid

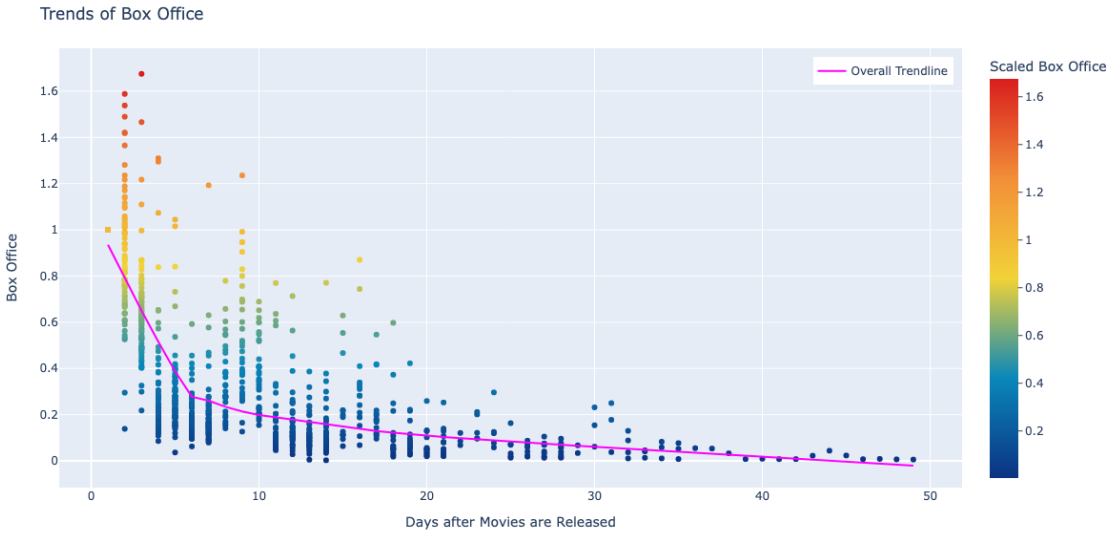


Figure 8: Trend of Box Office from 2020 to 2022

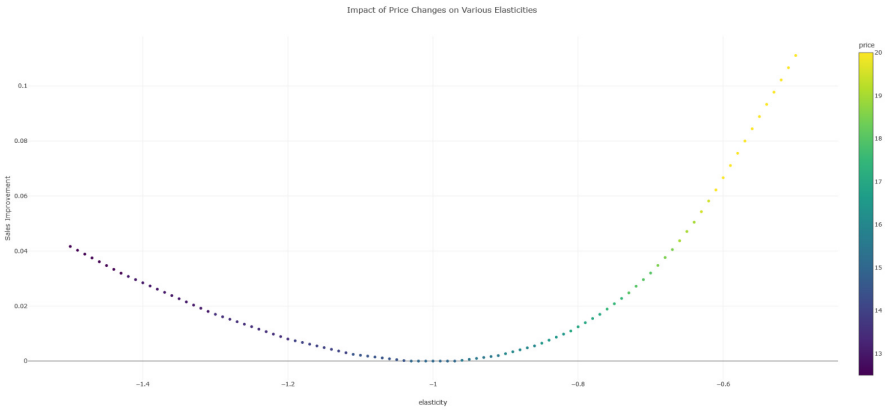


Figure 9: Pricing Analysis for Various Elasticities. Pricing Elasticity is the dependent variable for which the optimal price is selected such that the sales improvement is maximized.

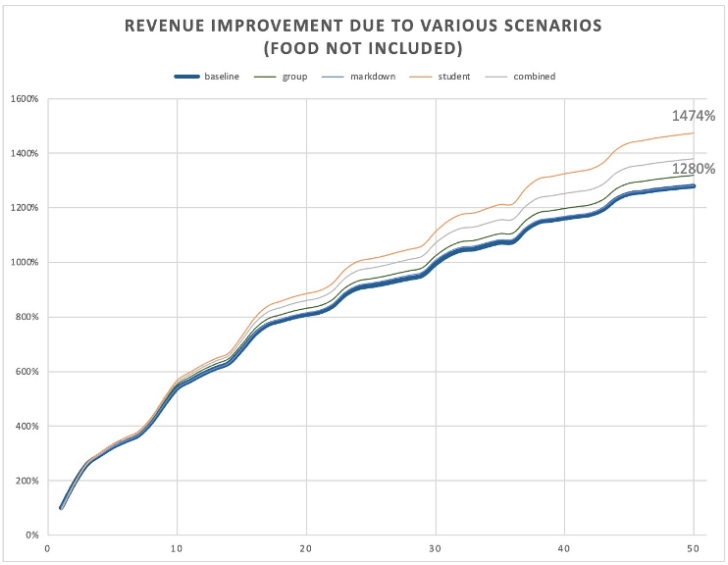


Figure 10: Cumulative Revenue Improvement from scenarios. Cumulative revenue based on first-day sales equal to 100%.

Financial highlights (in thousands of dollars, except theatre attendance in thousands of patrons and per share and per patron amounts)	Fourth Quarter			Full Year		
	2022	2021	Change (i)	2022	2021	Change (i)
Total revenues	$350,124	$299,951	16.7 %	$1,268,562	$656,669	93.2 %
Theatre attendance	9,208	10,245	-10.1 %	38,045	20,080	89.5 %
Net income (loss) (ii)	$10,168	$(21,778)	NM	$113	$(248,722)	NM
Net income (loss) as a percentage of sales (ii)	2.9 %	(7.3) %	10.2 %	— %	(37.9) %	NM
Cash provided by operating activities	$59,622	$27,480	117.0 %	$107,148	$61,004	75.6 %
Box office revenues per patron ("BPP") (iii)	$13.06	$12.29	6.3 %	$12.12	$11.77	3.0 %
Concession revenues per patron ("CPP") (iii)	$8.93	$7.49	19.2 %	$8.72	$7.93	10.0 %
Adjusted EBITDA (iii)	$74,186	$58,328	27.2 %	$251,694	$59,927	320.0 %
Adjusted EBITDAaL (ii) (iii)	$31,197	$20,198	54.5 %	$81,672	$(84,295)	NM
Adjusted EBITDAaL margin (ii) (iii)	8.9 %	6.7 %	2.2 %	6.4 %	(12.8) %	19.2 %
Adjusted free cash flow (iii)	$1,672	$(1,032)	NM	$3,339	$(151,517)	NM
Adjusted free cash flow per share (iii)	$0.026	$(0.016)	NM	$0.053	$(2.392)	NM
Earnings per share ("EPS") - basic and diluted (ii)	$0.16	$(0.34)	NM	$—	$(3.93)	NM

i. Period over period change calculated based on thousands of dollars except percentage and per share values. Changes in percentage amounts are calculated as 2022 value less 2021 value.

ii. 2022 includes expenses related to the Cineworld Transaction and associated litigation and claims recovery in the amount of $0.9 million (2021 - $2.3 million) for the fourth quarter and $3.6 million (2021 - $11.4 million) for the full year.

iii. Adjusted EBITDA, adjusted EBITDAaL, adjusted EBITDAaL margin, adjusted free cash flow per common share of Cineplex, BPP and CPP are measures that do not have a standardized meaning under generally accepted accounting principles ("GAAP"). These measures as well as other Non-GAAP other financial measures reported by Cineplex are defined in the 'Non-GAAP and Other Financial Measures' section at the end of this news release.

Figure 11: Cineplex fourth quarter and full year financial results 2020-2021 [Cineplex, 2023a]

	2022	2021
Revenues (note 19)		
Box office	$ 461,272	$ 236,320
Food service	381,386	186,998
Media	111,728	65,330
Amusement	246,601	134,473
Other	67,575	33,548
	1,268,562	656,669
Expenses and other income		
Film cost	238,897	114,674
Cost of food service	87,702	41,683
Depreciation - right-of-use assets	95,517	102,247
Depreciation and amortization - other assets	105,197	113,042
Gain on disposal of assets (notes 6, 9 and 19)	(57,807)	(28,283)
Other costs (note 20)	687,738	439,554
Share of loss of joint ventures and associates (note 9)	2,608	755
Interest expense - lease obligations (note 14)	61,842	58,590
Interest expense - other	60,826	65,138
Interest income	(277)	(232)
Foreign exchange	(1,371)	(43)
Loss (gain) on financial instruments recorded at fair value (note 15)	6,260	(8,790)
(Reversal) impairment of long-lived assets (note 11)	(19,880)	3,717
	1,267,252	902,052
Income (loss) before income taxes	1,310	(245,383)
Income tax expense (note 8)		
Current	1,197	3,339
Net income (loss)	$ 113	$ (248,722)
Net income (loss) per share - basic and diluted (note 21)	$ —	$ (3.93)

Figure 12: Cineplex consolidated statements of operations – Annual Report 2022 (expressed in thousands of Canadian dollars, except per share amounts) [Cineplex, 2023b]

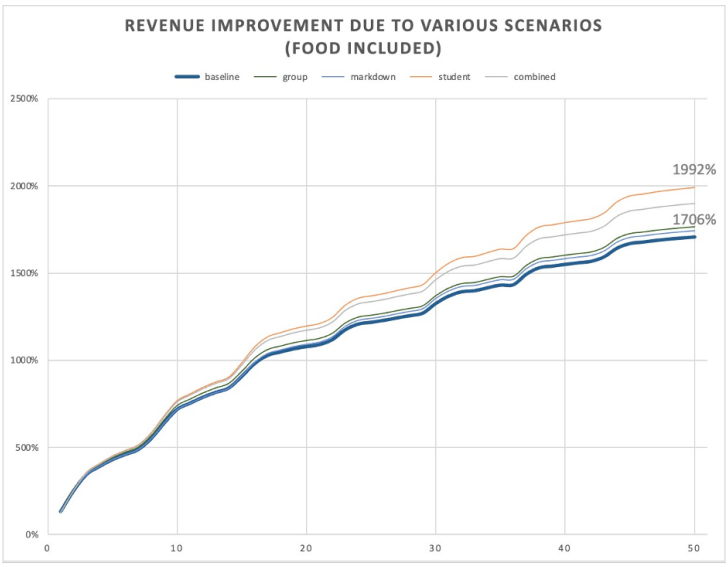


Figure 13: Cumulative revenue improvement with food considerations

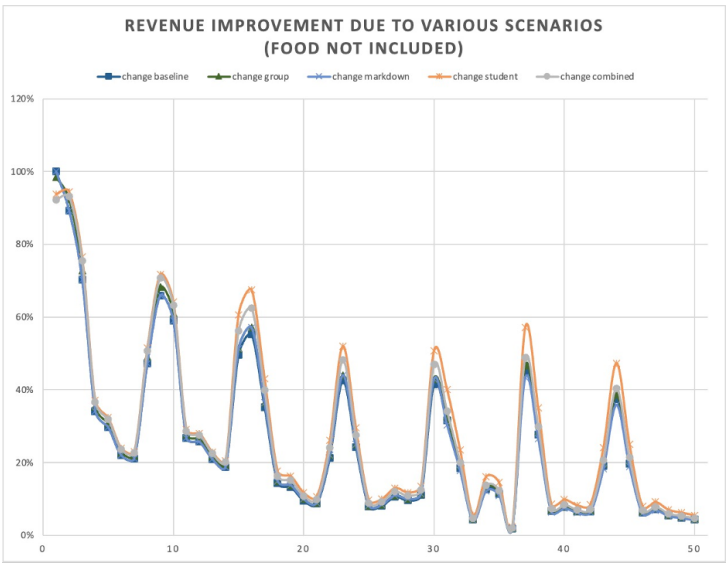


Figure 14: Daily Sales of movie tickets based on various strategies

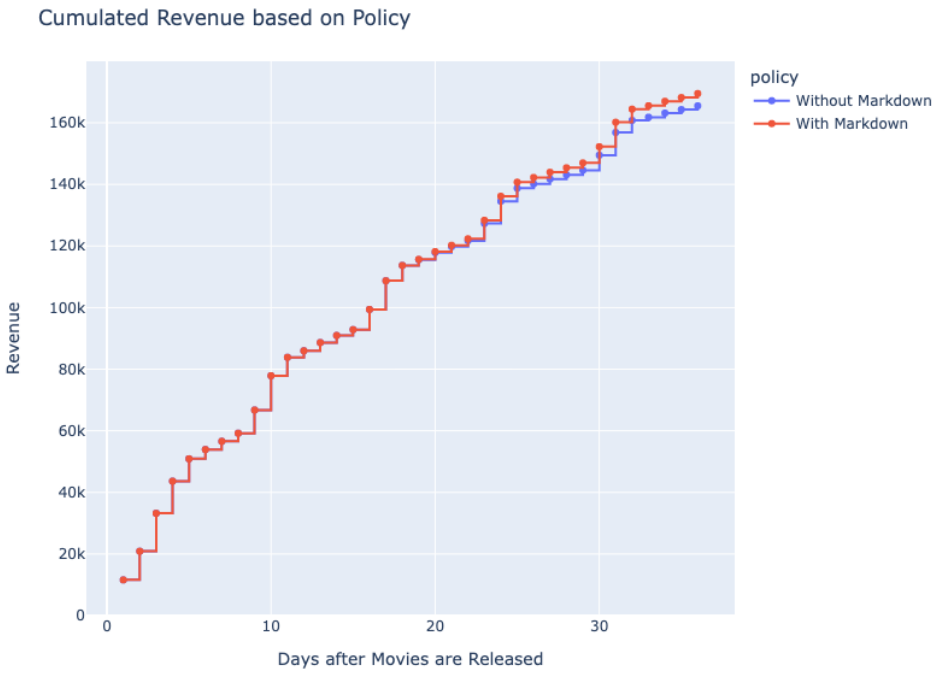


Figure 15: Cumulative Revenue based on Different Policies, using *Tenet* as an Example. (10% Discount with 20% Sales Increase)

6.2 Business Case

The executing of these strategies follows an experimentation approach conducted by a centralized data team. Executing group pricing, markdown strategies and student pricing will increase the volume of ticket sales and the revenue generated. The volume of consumers will primarily increase box office sales, but also, crucially, food and beverage purchases. These strategies address the challenge of increased views that have plagued movie theatres, particularly in the port-pandemic era. The modelling analysis demonstrates a revenue lift of at least 3%, with most of this coming in periods of historically lower attendance. The implementation should be carefully considered however, with larger chains, they can quickly implement one or all these strategies within the same geographies or outside of them. By experimenting and collecting data for 12 months on each strategy, cinemas will validate the effectiveness of each strategy. The experiment presents risks, if the strategies are not effective, then the resulting strategies will lose money around 10

7 References

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